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Summary

I investigate how software engineering students engage with supplementary learning materials and develop data-driven adaptive learning systems that respond to individual knowledge states and learning needs. My work spans software engineering education, learning analytics, and educational technology, producing both empirical insights and deployable tools that support large-scale, evidence-based teaching.

Education

- 2023 – present 📖 **Ph.D. Candidate in Computer Science, University of Alabama**
Advisor: Prof. Jeffrey Carver
Thesis: *Learning Beyond the Lecture: Rethinking Software Engineering Education with Supplementary Materials in the Generative AI Era*
- 2023 – 2025 📖 **M.S. in Computer Science, University of Alabama**
Comprehensive Exam: *Effective Methods for Teaching Software Engineering*
- 2020 – 2023 📖 **M.Sc. by Research in Computing Science, University of the South Pacific**
Advisor: Dr. Vani Vasudevan
Thesis: *Intelligent Education Management Information System (IEMIS) for Fiji*
- 2004 – 2008 📖 **B.Ed in Computer Science, University of Ibadan**
Minor: Mathematics
Graduated First Class; Best Graduating Student, Department of Computer Science

Research Publications

Journal Articles

- 1 **L. Badru** and J. Carver, “Personalizing supplementary materials in software engineering: Impact of preferences, learning styles, and gender,” *ACM Transactions on Computing Education*, vol. 26, no. 4, 2026. [DOI: 10.1145/3806053](#)
- 2 **L. Badru**, V. Vasudevan, M. Khan, and G. Lingam, “Mern stack web-based education management information systems for pacific island countries,” *SN Computer Science*, vol. 4, no. 1, p. 70, 2022. [DOI: 10.1007/s42979-022-01437-0](#)
- 3 **L. Badru**, V. Vasudevan, M. Khan, and G. Lingam, “State of education management information system (emis) in pacific melanesia small island developing states (sids),” *Directions: Journal of Educational Studies*, vol. 35, no. 2, pp. 107–118, 2021.

Conference Proceedings

- 1 **L. Badru**, M. An, J. Stamper, and J. Carver, “Optimizing learning: A comparative study of adaptive experiments and randomization in a software engineering course,” in *Proceedings of the Sixth Annual Workshop on A/B Testing and Platform-Enabled Learning Engineering (PELE), Learning@Scale 2025*, Palermo, Italy, 2025.

Preprints

- 1 **L. Badru** and J. Carver, *How undergraduate software engineering students navigate learning materials across contexts*, Under review at *ACM Transactions on Computing Education*, 2026.

- 2 L. Badru and J. Carver, *Supplementary learning materials in software engineering education: A systematic literature review*, Under review at SIGCSE Technical Symposium, 2026.

Work-in-progress

- 1 L. Badru and J. Carver, *Personalized support: Evaluating a digital platform for supplementary materials in software engineering education*, Work in progress, 2026.

Employment History

- 2023 – present  **Research Assistant**, Department of Computer Science, University of Alabama
- Conduct research in software engineering education, learning analytics, and adaptive learning systems.
 - Support multi-institutional validation studies of an AI-driven adaptive learning platform.
 - Contribute to data analysis, experimental design, and preparation of manuscripts for ACM and IEEE venues.
- 2016 – 2022  **Senior Education Officer / Acting Head, State EMIS**
Ministry of Education, Science and Technology, Abeokuta, Nigeria
- Led the design, development, and maintenance of the Education Management Information System (EMIS) serving 1,500+ schools and 500,000+ students.
 - Developed automated analytics tools, including a statewide Result Analyzer, improving data quality, reporting accuracy, and operational efficiency.
 - Served as a UNICEF Master Trainer, training 1,000+ personnel on school records management and annual census processes.
 - Contributed to state-level educational planning, including the Education Sector Plan (2017–2030) and related policy initiatives.
- 2016 – 2019  **Assistant Lecturer (Secondment)**, D.S. Adegbenro ICT Polytechnic, Ogun State, Nigeria
- Taught undergraduate courses in computing, mathematics, and information systems.
 - Designed assessments, practical exercises, and instructional materials for large student cohorts.
- 2014 – 2016  **Education Officer (Secondment)**, Staff Development Center, Bureau of Establishment & Training, Nigeria
- Delivered training in computer applications and digital literacy for adult learners and administrative staff.
- 2010 – 2014  **Master Grade II / Program Analyst**, Teaching Service Commission, Abeokuta, Nigeria
- Provided technical support for educational data systems and contributed to ICT capacity-building initiatives.

Teaching Experience

- 2026

Workshop Facilitator, Tai Solarin Federal University of Education, Ijagun, Ogun State, Nigeria
- Designed and led a two-day virtual workshop on *Agentic Coding* for advanced Computer Science and Software Engineering students.
 - Created interactive activities and guided practice sessions to reinforce key concepts.
- 2016 – 2019

Assistant Lecturer, D.S. Adegbenro ICT Polytechnic, Ogun State, Nigeria
- Taught undergraduate courses in computing, mathematics, and information systems to cohorts of up to 120 students.
 - Designed and delivered lectures, tutorials, assessments, and practical exercises.
 - Employed worked examples, structured progression, and short problem-solving tasks to support student understanding.
- 2014 – 2016

Education Officer, Staff Development Center, Bureau of Establishment & Training, Nigeria
- Delivered training in computer applications and digital literacy for adult learners and administrative staff.
 - Developed instructional materials and hands-on exercises tailored to diverse learner backgrounds.
- 2010 – 2026

Mentorship and Student Support
- Mentored over 100 high school and college students, supporting academic growth and career development.
 - Provided guidance on computing fundamentals, project work, and professional pathways.

Skills

Programming Languages	Python, JavaScript, PHP
Frameworks & Platforms	React, Node.js, Express, MERN Stack, MOOClet Framework (Adaptive Learning)
Artificial Intelligence	OpenAI API, Prompt Engineering, AI-generated Assessments, ChatGPT-based Content Creation, Design of AI-driven Adaptive Learning Systems
Learning Analytics	Data-driven Personalization, Predictive Analytics, Behaviour Modelling, Clickstream & Engagement Analysis, A/B Testing for Educational Content
Educational Technology	Development of Adaptive Experimentation Platforms, Learning Tool Integration, Technology-enhanced Teaching Methods
Research Methods	Experimental Design, Mixed-Methods Research, Survey Design & Analysis, Statistical Modelling, Multi-institution Validation Studies
Software Engineering Education	Curriculum Design, Instructional Material Development, Pedagogical Strategies for SE Courses, Evaluation of Supplementary Learning Resources

Skills (continued)

Data & Research Tools	■ Alteryx, Tableau, Pandas, NumPy, SciPy, Statistical Analysis of Survey & Learning Data
Databases	■ MySQL, PostgreSQL, MongoDB
Teaching & Facilitation	■ Lecture Design, Assessment Development, Worked Examples, Structured Practice, Problem-based Learning, Agentic Coding Workshops, Teaching Large Classes (up to 120 students)

Miscellaneous Experience

Scholarly Contribution

- 2022 ■ **Reviewer**, *Education and Information Technologies* (Springer Nature). Reviewed manuscripts in educational technology and learning analytics.

Awards and Scholarships

- 2021 ■ **Research Grant**, Queen Elizabeth Commonwealth Scholarship, United Kingdom (3,000 USD)
- 2020 – 2021 ■ **Commonwealth Scholarship**, United Kingdom (value over 50,000 USD), University of the South Pacific, Fiji
- 2019 ■ **Dean's Award for Academic Excellence & Global Leader Scholarship**, University of Bath, United Kingdom (value £10,000)
- 2009 ■ **Best Graduating Student**, Department of Computer Science, Tai Solarin University of Education, Nigeria (value 500 USD)

Certifications

- 2021 ■ **Predictive Analytics for Business**, Udacity (Alteryx & Tableau)
- **Microsoft Azure Fundamentals**, Microsoft
- **Ideas for a Better World: Leading Change Through Policymaking**, British Council

Mentorship and Service

- 2010 – 2026 ■ Advised and mentored over 100 senior secondary and university students, supporting academic and professional development.
- 2026 ■ **Judge**, Alabama Robotics Contest, University of Alabama. Evaluated robot navigation accuracy, task completion, and engineering design.
- 2024 – 2026 ■ **Volunteer**, Alabama Robotics Contest, University of Alabama. Supported event logistics and prepared the robot navigation field.
- 2017 – 2019 ■ **UNICEF Master Trainer** on school records management and the Annual School Census (ASC) process.
- 2017 ■ Contributed to the development of the State Education Sector Operational Plan (SESOP 2017–2021) and State Education Sector Plan (SESP 2017–2030).
- 2016 ■ Served as a technical expert for the Education Policy & Strategy Simulation (EPSSim) initiative.

Professional Memberships

- **Association for Computing Machinery (ACM)**
- **Institute of Electrical and Electronic Engineers (IEEE)**
- **Special Interest Group on Computer Science Education (SIGCSE)**

References

Jeffrey Carver, PhD

James R. Cudworth Professor
Associate Department Head for Graduate Studies
Department of Computer Science
University of Alabama

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Relationship: Academic Advisor

Jeff Gray, Ph.D.

University Distinguished Professor
Department of Computer Science
University of Alabama
Director, Randall Research Scholars Program

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Relationship: Thesis Committee Member

John Stamper, PhD

Associate Professor
METALS Program Director
Carnegie Mellon University

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Relationship: Thesis Committee Member